PROJECT REPORT

Auto Ticket Classification using Flow Designer

Team Members

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ServiceNow • Flow Designer • Incident Management

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1. INTRODUCTION

1.1 Project Overview

Auto Ticket Classification using Flow Designer is a ServiceNow-based automation project designed to automatically classify IT support incidents based on the issue reported by a user. In a school environment, the IT helpdesk receives multiple incident requests from students and teachers related to Wi-Fi connectivity, projector failures, password problems, computer performance, hardware failures, account access and other technical issues.

Traditionally, IT staff manually review each incident and select an appropriate Category and Subcategory. This process can be repetitive, time-consuming and prone to inconsistent classification. The proposed solution uses ServiceNow Flow Designer to automate the classification process by evaluating information in the incident's Short Description and Description fields.

1.2 Purpose

- Automatically identify the type of IT issue.
- Assign the appropriate Category and Subcategory.
- Reduce repetitive manual work for IT support staff.
- Improve ticket routing and processing efficiency.
- Provide consistent incident classification.
- Send automated notifications to users.
- Provide a simple and maintainable low-code/no-code automation solution.

2. IDEATION PHASE

2.1 Problem Statement

The school IT helpdesk receives multiple support requests every day from students and teachers. Common incidents include Wi-Fi connectivity issues, projector failures, password reset requests, computer performance problems, hardware failures, account access problems and network issues.

Currently, IT staff manually review each ticket, identify the issue type, and select the correct category and subcategory. As the number of requests increases, this manual process becomes difficult to scale and can delay ticket routing.

2.2 Empathy Map Canvas

Empathy Area	School IT Helpdesk Staff
Says	“There are many tickets to review every day.”
Thinks	“Manual classification takes too much time.”
Does	Reviews ticket descriptions and assigns categories manually.
Feels	Overloaded when incident volume increases.
Pain Points	Manual classification, repetitive work, inconsistent categorization and delayed routing.
Needs	Automatic classification, consistent categorization and faster ticket processing.

2.3 Brainstorming

Several possible approaches were considered during ideation:

- Manual classification by IT staff.
- Classification using ServiceNow Flow Designer.
- Machine-learning-based classification.
- Keyword and rule-based classification.
- Classification using an external AI service.

ServiceNow Flow Designer was selected because it provides a visual, low-code approach and can directly automate ServiceNow incident records without requiring a large custom application.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	User Action	System Activity
1. Identify Problem	Student/teacher faces an IT issue.	No system activity yet.
2. Create Ticket	User submits an incident.	Incident record is created.
3. Analyze Ticket	User waits for processing.	Flow reads incident information.
4. Classification	No manual category selection required.	Category and Subcategory are assigned.
5. Notification	User receives confirmation.	Automated email is triggered.
6. Support	IT team handles the issue.	Correctly classified ticket is available.
7. Resolution	Issue is resolved.	Incident is updated and closed.

3.2 Solution Requirement

Functional Requirements

- Automatically classify IT tickets based on the issue description.
- Assign both Category and Subcategory without manual intervention.
- Support dependent choice logic between Category and Subcategory.
- Analyze the Short Description and Description fields.
- Send an automated email notification to the caller after ticket creation.
- Store and update incident information in ServiceNow.
- Reduce the need for manual ticket classification.
- Provide an easy-to-maintain automation flow.

Non-Functional Requirements

- Reliable and consistent behavior.
- Fast processing after ticket creation.
- Maintainable flow and conditions.
- Scalability for increasing ticket volumes.
- Minimal manual intervention.
- Compatibility with the ServiceNow platform.

3.3 Data Flow Diagram

High-level data flow: Student / Teacher → Create Incident → ServiceNow Incident Table → Flow Designer → Classification Logic → Category + Subcategory → Update Incident → IT Helpdesk

The system receives incident information from the user. Flow Designer processes the incident and checks predefined conditions. When a matching condition is found, the appropriate Category and Subcategory are assigned to the incident record.

3.4 Technology Stack

Technology	Purpose
ServiceNow	IT Service Management platform.
Flow Designer	Automation and workflow implementation.
Incident Management	Ticket creation and management.
ServiceNow Tables	Storage of incident information.
Conditions / Flow Logic	Classification rules.
Email Notifications	Automated caller notification.
ServiceNow UI	User interaction and ticket management.

4. PROJECT DESIGN

4.1 Problem Solution Fit

Existing Problem	Proposed Solution
Manual ticket classification	Automatic classification through Flow Designer.
Time-consuming process	Workflow automation.
Incorrect or inconsistent category selection	Predefined classification conditions.
Manual subcategory selection	Automatic dependent subcategory selection.
Delayed ticket routing	Immediate classification after creation.
Repetitive IT staff work	Automated workflow.
Manual notification	Automated email notification.

4.2 Proposed Solution

The proposed solution uses ServiceNow Flow Designer to automate incident classification. When a new incident is created, the flow reads the Short Description and Description, evaluates predefined conditions, assigns the appropriate Category and Subcategory, updates the incident, and sends an automated notification to the caller.

Working Process

- A student or teacher creates an IT incident.
- The incident contains a Short Description and Description.
- The Flow Designer flow is triggered when the incident is created.
- The flow reads the incident information.
- Predefined conditions and keywords are evaluated.

- The appropriate Category is identified.
- The corresponding Subcategory is selected.
- The incident record is updated.
- An automated email notification is sent to the caller.
- The classified incident becomes available to the IT support team.

4.3 Solution Architecture

Architecture Flow

Student / Teacher
↓
Create Incident Ticket
↓
ServiceNow Incident Table
↓
Flow Designer
↓
Read Short Description + Description
↓
Classification Logic
↓
Category + Subcategory
↓
Update Incident
↓
Email Notification
↓
IT Helpdesk

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Phase	Activity
Phase 1	Requirement Analysis
Phase 2	Problem Identification
Phase 3	Brainstorming
Phase 4	ServiceNow Environment Setup
Phase 5	Incident Configuration
Phase 6	Flow Designer Development
Phase 7	Category/Subcategory Classification
Phase 8	Email Notification Configuration
Phase 9	Functional Testing
Phase 10	Performance Testing
Phase 11	Result Analysis
Phase 12	Documentation

The project follows an iterative development approach: Requirement → Design → Development → Testing → Modification → Final Implementation.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

The system was tested using different types of IT incidents to verify automatic classification. The examples below represent typical scenarios used to validate the workflow.

Test Case	Scenario	Expected Classification	Status
TC01	Wi-Fi issue	Network → Wi-Fi	Passed
TC02	Projector failure	Hardware → Projector	Passed
TC03	Password problem	Access → Password	Passed
TC04	Slow computer	System → Performance	Passed
TC05	Network connectivity	Network → Connectivity	Passed
TC06	General hardware issue	Hardware → Device	Passed

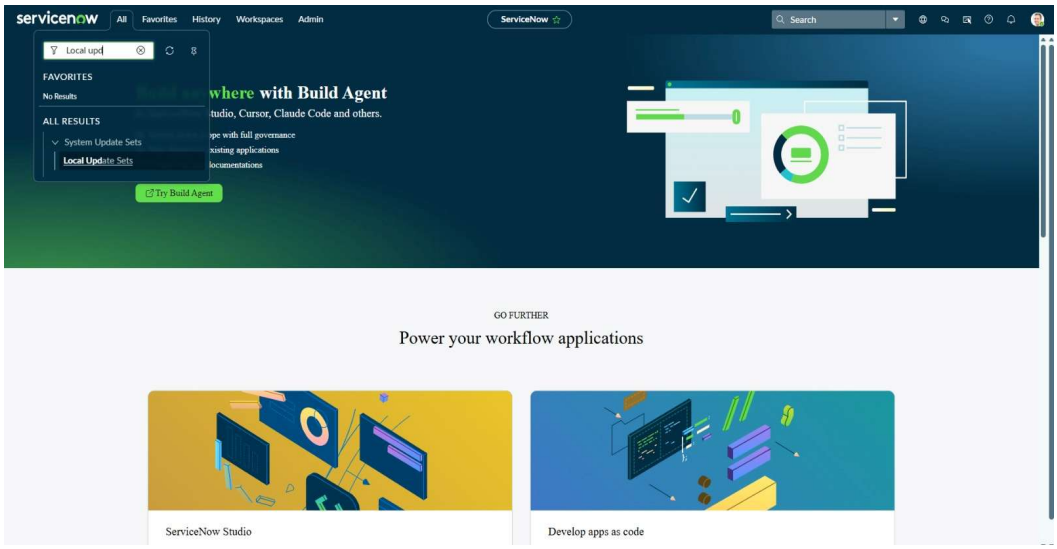
Performance Observation

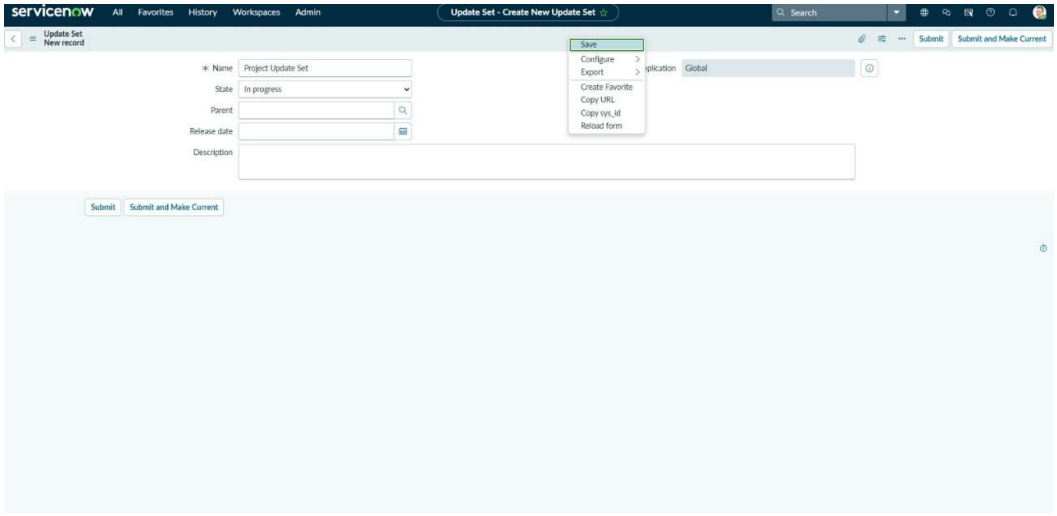
The flow processes an incident automatically after creation and removes the need for IT staff to manually classify every ticket. This improves processing efficiency and promotes consistent classification.

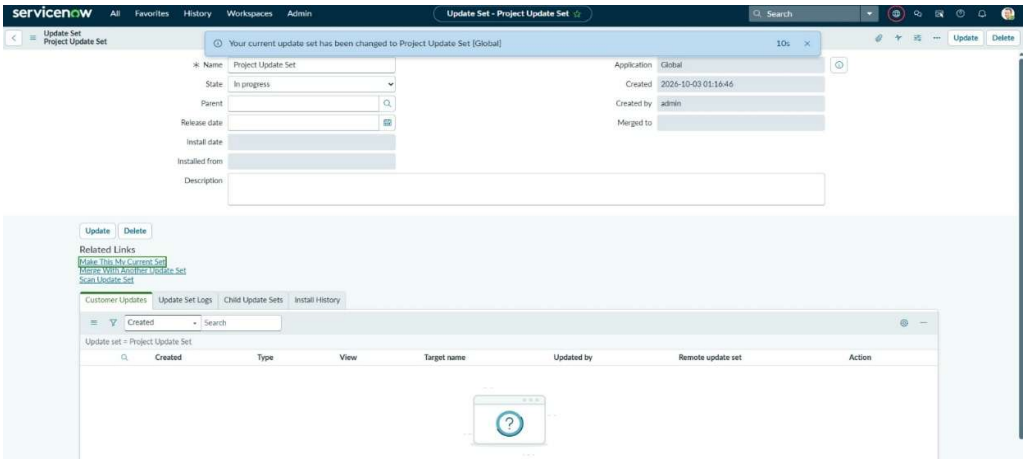
7. RESULTS

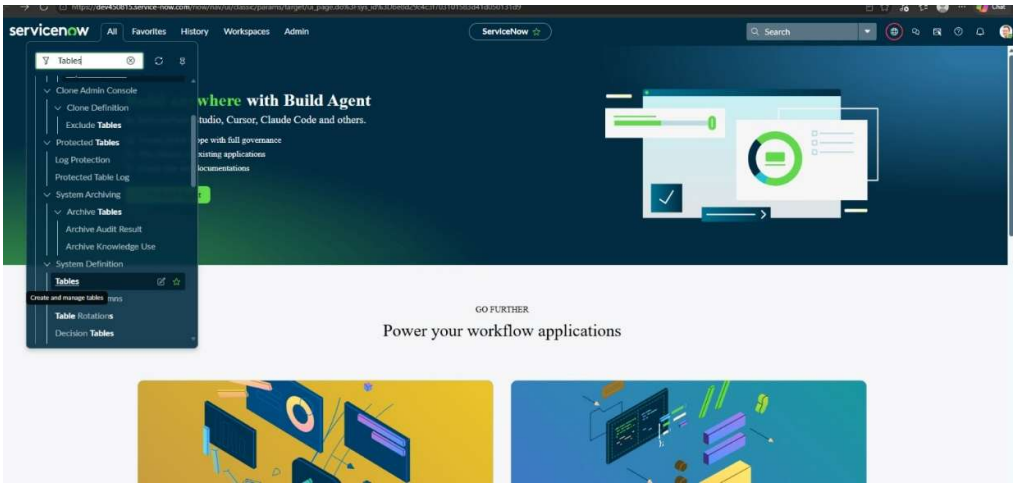
7.1 Output Screenshots

The following screenshots document the project concept, report format, problem statement, objective and initial business requirements.









Final Result :

The project demonstrates an automated approach for classifying IT incidents using ServiceNow Flow Designer. By reducing manual classification, the solution supports faster and more consistent ticket processing.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Reduces manual work for IT support staff.
- Faster ticket processing after incident creation.
- Improves consistency of classification.
- Easy visual maintenance using Flow Designer.
- Supports better ticket routing.
- Automated caller notifications.
- Can scale with increasing incident volume.
- Requires little or no traditional application code.

Disadvantages

- Keyword-based classification may not understand unusual descriptions.
- Similar words may occasionally result in incorrect classification.
- New issue types may require additional conditions.
- The solution depends on correct ServiceNow configuration.
- Complex classification requirements may require advanced automation or AI.
- Incomplete ticket descriptions can reduce classification accuracy.

9. CONCLUSION

The Auto Ticket Classification using Flow Designer project demonstrates how ServiceNow can automate repetitive IT service management activities. The traditional manual process requires IT staff to review every incident and select the appropriate category and subcategory, which can consume time and introduce inconsistency.

The proposed Flow Designer solution evaluates incident information and automatically assigns the appropriate classification based on predefined conditions. It can also support automated notifications and a consistent workflow for IT support.

Overall, the project shows that low-code workflow automation can reduce manual effort, improve ticket processing efficiency and provide a better support experience for students, teachers and IT staff.

10. FUTURE SCOPE

The current rule-based approach can be extended with advanced automation and Artificial Intelligence to improve understanding of natural-language incident descriptions.

- AI-based ticket classification instead of simple keyword rules.
- Natural Language Processing to understand the meaning of descriptions.
- Automatic assignment-group selection.
- Automatic priority prediction.
- Sentiment analysis for urgent or frustrated users.

- Duplicate ticket detection.
- AI-powered suggested resolutions using historical incidents.
- Multi-language ticket support.
- Analytics dashboards for ticket volume and resolution time.
- Predictive incident management using historical ticket data.

11. APPENDIX

Source Code

The project primarily uses ServiceNow Flow Designer, so the implementation is mainly configuration and flow-based rather than a traditional standalone source-code application.

Source Code / Flow Configuration: To be added from the team's ServiceNow instance.

Dataset Link

The project does not require a conventional external dataset. Ticket classification is performed using predefined conditions and incident information within ServiceNow.

Dataset: Not Applicable / ServiceNow Incident Records

GitHub & Project Demo Link

GitHub Repository: _____

Project Demo: _____

Submission Note: Replace the GitHub/Demo placeholders with the actual links before submission. If the ServiceNow instance uses different Category/Subcategory names, update the testing examples to match the actual configuration.